

Technical Notes: Reference Management and Study Selection

1. Tool Overview

This systematic literature review used Zotero for reference management, archival organization, and bibliography preparation. Rayyan AI was used to support deduplication, screening organization, and reviewer coordination. These tools helped maintain a transparent audit trail from initial record identification to final study inclusion. No records were excluded solely by automation; all exclusion decisions were reviewed and confirmed by the authors.

2. Multi-Stage Selection Workflow

The selection process was divided into identification, database filtering, deduplication, title and abstract screening, full-text eligibility assessment, and quality assessment:

- **Stage 1 & 2 (Identification):** The initial electronic search across five databases yielded 639 records: Web of Science (n = 181), Scopus (n = 174), ScienceDirect (n = 120), IEEE Xplore (n = 116), and ACM Digital Library (n = 48).
- **Stage 2: (Database Filtering):** Predefined database filters were applied to restrict the scope to English-language, peer-reviewed journal articles published between 2021 and 2025. This resulted in the exclusion of **242** records, leaving **397** records.
- **Stage 3 (Deduplication):** The remaining records were consolidated in Zotero and imported into Rayyan AI. Deduplication identified and removed 47 overlapping records, leaving **350 unique records** for title and abstract screening.
- **Stage 4: Title and Abstract Screening** Two independent reviewers screened the titles and abstracts of the 350 records. This phase led to the exclusion of **226** records, consisting of 14 secondary studies, such as surveys and reviews, and **212** records that were irrelevant to the technical scope of ML-based insider threat detection, datasets, preprocessing, or feature engineering. This left **124** records for full-text retrieval..
- **Stage 5: Full-Text Eligibility and Quality Assessment (QA)** Of the **124** records sought for retrieval, 7 full-text reports could not be retrieved in full-text, leaving **117** studies for eligibility assessment. After applying the inclusion/exclusion criteria and Quality Assessment (QA) framework, 7 studies were excluded, and **110** primary studies were included in the final review.

3. Justification for Final Exclusions

During the full-text eligibility phase, **7 studies** were excluded for failing to meet the QA threshold. Specifically:

- **Three (3) papers** were excluded because they did not provide sufficient technical detail on the datasets, preprocessing steps.
- **Four (4) papers** Four papers were excluded because their research scope was outside the specific focus of ML-based insider threat detection.

4. Data Integrity and Peer Review

To minimize selection bias, the selection process was conducted by two independent reviewers. The reviewers carried out title and abstract screening, full-text eligibility assessment, and QA scoring. Any disagreements were resolved through shared screening notes and discussion. The complete process is summarized in the PRISMA 2020 flow diagram included in the manuscript.